



Breathing Cycle Detection for Respiratory Tele-health Systems

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Abstract

Recent advancements in wearable devices have enabled the acquisition of lung sounds in real time. By analyzing these signals, key indicators such as respiratory cycles and heart sound components can be extracted, hence enabling the development of tele-health solutions for remote assessment of pulmonary conditions. Particularly, detecting respiratory cycles within the collected sound data plays a crucial role in both clinical and diagnostic applications. Accurate identification of breathing patterns facilitates the assessment of respiratory function and supports early detection of anomalies, including COPD, pneumonia, asthma, and COVID-19. In this paper, we promote a two-step process that first estimates breathing sound signal envelope (in time domain) and then analyzes the envelope peaks/valleys to calculate the respiratory cycle. Three methods that follow such a process are proposed. We examine the practicality, scalability, and efficacy of these methods in both healthy and pathological cases, highlighting their potential for integration into real-world respiratory monitoring and screening systems. We further evaluate their performance using the public dataset ICBHI-2017, which allows comparative analysis of various respiratory conditions.

Keywords Tele-health · Respiratory monitoring · Breathing cycle analysis · Signal envelope detection

1 Introduction

The major advances in microelectronics and wireless communication technology are transforming societies and enabling unconventional services in almost all application domains. In particular, the healthcare industry has been adopting a service model that increases coverage and improves resource utilization,

where patients are examined and their conditions are monitored remotely. Such a service model is often referred to as tele-health and brings about multiple advantages, including (i) offering the convenience of at-home recovery, (ii) reducing the rate of hospital admission and, in turn, the cost of health care, (iii) enabling around the clock health assessment, and (iv) limiting the load on healthcare professionals. The viability of tele-health solutions has become apparent since the beginning of the COVID-19 pandemic, where the rate of infection grew very high and the healthcare professionals were subjected to great risk. Continual monitoring of patients, especially those suffering from chronic diseases, allows them to live a normal life and enables timely intervention as soon as need arises.

Remote Respiratory Monitoring: As highlighted in Fig. 1, tele-health could be a viable option for dealing with lung diseases. Respiratory illnesses are conditions that affect the lungs and airways, impairing normal breathing. They encompass a variety of disorders, from acute infections such as pneumonia and bronchitis to chronic illnesses, such as asthma, chronic obstructive pulmonary disease (COPD), and lung cancer. These chronic conditions require frequent patient hospitalization, and depending on their severity level,

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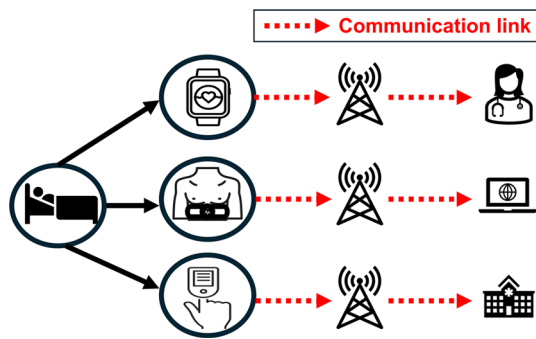


Fig. 1 An articulation of systems for remote respiratory monitoring

they would need immediate intervention by a healthcare provider. Managing the care of patients with chronic respiratory diseases imposes a major burden in terms of facilities and cost on the healthcare system, especially knowing that COPD and asthma affect approximately 213 million [1] and 262 million people [2] worldwide, respectively, and that acute respiratory distress syndrome (ARDS) affects over 3 million people each year [3]. By remotely monitoring the health conditions of these patients, appropriate decisions may be made and a significant reduction in hospital visits can be achieved.

Similar benefits of tele-health monitoring technology could be noted for handling infectious, non-chronic respiratory diseases that can be caused by viruses, bacteria, environmental pollutants, or underlying health conditions. By remotely treating and following up with patients, widespread of communicable diseases could be prevented and vulnerable people could be protected. Each year, seasonal influenza infects approximately 1 billion people, resulting in up to 650,000 deaths [4]. In addition, respiratory syncytial virus (RSV) causes approximately 3.6 million hospitalizations and 100,000 deaths annually in children under five years of age, with nearly half of the deaths occurring in infants under six months, predominantly in low- and middle-income countries [5]. Despite seasonal fluctuations, these illnesses continue to pose significant challenges to public health and healthcare systems. The COVID-19 pandemic was a stark example, bringing the world to a standstill a few years ago due to its unprecedented severity and rapid spread.

Breathing Cycle Detecting: The onset and progression of the aforementioned respiratory diseases can be mitigated through regular monitoring and timely interventions. The tele-health technology could automate the monitoring process and the evaluation of patient's conditions. A wide range of wearable devices can capture and transmit data relevant to respiratory function, enabling continuous and remote assessment. The underlying approach is to use a digital stethoscope to capture the chest sound and complement it with additional physiological measurements that reflect breathing conditions; all measurements are then analyzed

to detect irregularities that indicate airflow obstruction. The breathing cycle is typically the focus, regardless of how the analysis is conducted. The length of the breathing cycle is indicative to how smoothly air flows during the patient's inhale and exhale. While doctors are trained to determine the breathing cycle based on the chest sound, special techniques are needed to do the same in a tele-health system.

Detecting a respiratory cycle is quite challenging. Typically, the sound signal collected by a digital stethoscope includes multiple breathing cycles. To conduct health assessment, such a signal needs to be segmented based on the boundary of the individual cycles. The boundary of a cycle is marked by two prominent features, often referred to as S_1 and S_2 , reflecting the heart sound marking the beginning and end of systole, respectively. In a tele-health system, automatic detection of S_1 and S_2 is complicated by: (i) the presence of ambient noise during data collection, (ii) breathing irregularities, and (iii) abnormalities such as coughs, wheezing, rhonchi, crackles, etc. Signal processing and machine learning (ML), especially deep learning (DL) methods are the prime candidates for tackling this complication.

Contribution: This paper studies the effectiveness of signal processing methods in determining the breathing cycle in a chest sound recording. Unlike ML and DL models, signal processing methods do not require prior knowledge or large amounts of annotated data and are thus deemed suitable for applications with limited data availability. These methods are typically deterministic and transparent, providing explainable mechanisms for detection and analysis. In addition, signal processing techniques are computationally lightweight, making them robust and ideal for use on low-power wearable devices. Although ML/DL models are powerful, they are computationally intensive, less interpretable, and highly dependent on the quality and volume of training data.

Existing work using signal processing techniques has focused on respiratory rate rather than cycle and consequently does not yield sufficiently high accuracy. The paper particularly promotes a two-step process involving a combination of signal processing techniques, where the breathing sound signal envelope is first determined in time domain, and then the peaks/valleys of the envelope are analyzed to calculate the respiratory cycle. Three methods that follow this process are studied in the paper, specifically, (1) a bandpass filtering and peak detection-based algorithm, (2) a spectrogram and envelope-based peak detection algorithm, and (3) an ensemble empirical mode decomposition (EEMD) and valley detection-based algorithm. We conduct an empirical assessment to compare their performance and provide guidelines for practical use. For that, we utilize the publicly available ICBHI-2017 dataset [6], which contains a combination of breathing sound signals collected from both healthy individuals and patients with respiratory conditions

such as COPD, pneumonia, asthma, bronchiectasis, pneumonia, and bronchiolitis. We have selected a subset of healthy and unhealthy samples. We evaluate the robustness, time efficiency, and effectiveness of the proposed methods. We use several metrics that are commonly employed for breathing pattern analysis, such as the root-mean-square error (RMSE), the mean absolute error (MAE), and mean absolute percentage error (MAPE).

The remainder of the paper is organized as follows. The next section discusses the various sensor modalities that are used to assess chest conditions, highlighting the popularity of acoustics. Section 3 covers related work. The proposed respiratory cycle detection process is detailed along with the three proposed methods in Sect. 4. Section 5 reports the performance results. Finally, the paper is concluded in Sect. 6.

2 Wearable Respiratory Sensors

This section provides an overview of the relevant sensor modalities for respiratory monitoring and highlights the focus of the paper.

Sensing Modalities: As noted earlier, the incorporation of wearable devices has major economic and social advantages. In the context of respiratory illness, they enable real-time and noninvasive monitoring, making health care accessible to people across all segments. Therefore, wearable device—driven solutions have the potential to significantly reduce deaths caused by the deterioration of respiratory conditions, particularly in low- and middle-income countries. There is a wide range of sensing modalities for lung health monitoring that can be employed on a wearable device. For example, photoplethysmography (PPG) signals, obtained from smartwatches or finger rings, provide blood volume changes used to estimate oxygen saturation (SpO_2), heart rate, and respiratory rate [7]. In addition, respiratory inductance belts measure thoracic and abdominal movements to extract respiratory rate and tidal volume. These measurements can be complemented by digital stethoscope data such as phonocardiograms (PCG) [8] and accelerometer signals [9].

Respiratory monitoring has been instrumental in the diagnosis and management of various diseases such as COPD, pneumonia, asthma, and bradycardia. There are several approaches to continuous respiratory monitoring, which can be broadly categorized into two forms: (1) signal-based methods (e.g., PPG, EKG, ECG, and PCG) and (2) image-based methods (e.g., CT scan, MRI, X-ray) [10]. Image-based techniques typically require extensive resources for data acquisition and processing. In contrast, signals acquired via digital stethoscopes and wearable devices offer a more economical and scalable solution for daily health monitoring, especially for individuals in low- and middle-income regions.

Digital stethoscope: This paper focuses on the use of signal processing methods to analyze breathing patterns in data obtained from digital stethoscopes. A digital stethoscope is a medical device that electronically captures, amplifies, and digitizes body sounds such as heartbeats, lung sounds, and bowel noises. Unlike traditional stethoscopes that rely on acoustic transmission, digital stethoscopes convert sound waves into electronic signals for enhanced listening, visualization, and analysis. These signals can be filtered, recorded and transmitted via Bluetooth or USB for remote consultation or integration with electronic health records. Digital stethoscopes are particularly valuable in tele-health, automated diagnostics, and continuous health monitoring. Several types of signals can be acquired using a digital stethoscope, including ECG (Electrocardiogram), EKG (Elektrokardiogramm), PPG, and PCG. Each of these signals is capable of providing valuable information on respiratory conditions.

From these physiological signals, various aspects of the breathing pattern can be derived, which serve as strong indicators of respiratory health. In healthy individuals, respiration is usually regular, with a rate of 12 to 20 breaths per minute [11] and balanced phases of inspiration and expiration. In contrast, patients with conditions such as COPD or ARDS often exhibit abnormal breathing patterns, including rapid, shallow breathing, irregular rhythms, or prolonged expiration. For example, a COPD patient may have a respiratory rate that exceeds 25 breaths per minute [12] with labored breathing, whereas conditions such as bradypnea can result in abnormally slow respiration [13].

Among the various biosignals that can be acquired from digital stethoscopes, PCG is widely used to analyze breathing patterns and detect anomalies in respiratory health. Typically, a PCG signal consists of multiple breathing cycles, each containing two prominent heart sound features: S_1 and S_2 . S_1 (“lub”) marks the closure of the mitral and tricuspid valves, signaling the beginning of systole, while S_2 (“dub”) corresponds to the closure of the aortic and pulmonary valves, marking the end of systole. The interval between two consecutive S_1 sounds can be considered as one respiratory cycle. Within a single respiratory cycle, several parameters can be derived, such as inhalation duration, exhalation duration, and pause duration. Figure 2 shows an example PCG signal and marks the observed S_1 and S_2 incidents. In this paper, we focus on detecting breathing cycles from PCG signals.

Example Respiratory Wearables: As noted above, there are numerous methods that can provide a quantitative assessment of respiratory health. Existing respiratory monitoring wearable devices can be classified based on the underlying sensing modality and on their applications. Measuring air flow, the volume of inhaled and exhaled air, and the time required for each cycle is quite popular. The air flow tracking device of [14] employs four air-coupled microphones (Sony ECM77) with a half-sphere coupling chamber, designed



Fig. 2 Showing S_1 and S_2 instances in one breathing cycle of a PCG signal

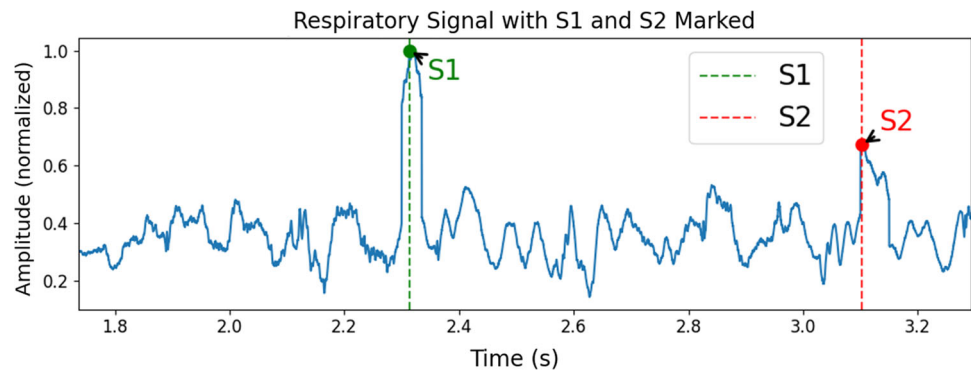
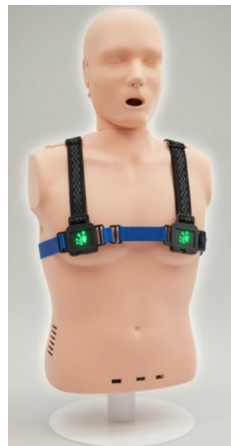


Fig. 3 Illustrating the use of WearME device applied through a strap to monitor the breathing regularity of a patient. Credit is due to LASARRUS Clinic and Research Center



to reduce high-frequency noise. The LASARRUS Wearable Multimodal Electroacoustic (WearME) system [15] is a bilateral respiratory monitoring system that functions as a digital stethoscope, temperature sensor and motion tracker. A patient can use the WearME device through a strap attached to the chest, as shown in Fig. 3. A WearME node tracks body posture and temperature and monitors heart and lung functions, detecting breathing anomalies in real time. Meanwhile, in [16] air flow is monitored using a pneumotachograph (Fleisch No. 3) connected to a pressure transducer.

The thoracic movement is a reliable indicator of respiratory health, and many chest belt-integrated devices are available on the market. This motion can be tracked using various types of sensors, including resistive, piezoresistive, and optical sensors embedded in chest belts. Examples of such a type of devices include StretchSenseTM fabric sensors [17], FlexiForce[®] piezoresistive sensors [18], electro-resistive bands from Adafruit [19], piezoresistive foil strain gauges such as the VISHAY SR-4 [20], and RespiWear, which employs optical sensor based Fiber Bragg Grating technology for precision strain sensing [21]. Optical sensors are not only used in chest belts but are also integrated into wearable smartwatches, enabling the measurement of PPG and SpO_2 . Numerous studies have investigated the use of PPG signals for estimating respiratory rates. Wrist-worn smart watches

provide this functionality, including the Apple Watch [22], Empatica EmbracePlus [23], and Whoop 4.0 [24].

While wearable watches and finger-based devices have shown promise, with several studies demonstrating effective respiratory rate detection using PPG signals [22, 24], chest belt devices remain the gold standard, offering more meticulous and precise measurements of respiratory parameters. Airflow and chest motion measurements represent fundamental modalities for respiratory monitoring. Beyond these, chest belts can be equipped with additional sensors to capture signals such as ECG, EKG, and PCG. When integrated within a single system, such multisensor chest belts operate as comprehensive monitoring platforms and can be classified as digital stethoscopes. Digital stethoscopes enable the assessment of chest movements using sensors such as accelerometers and strain gauges, while simultaneously acquiring cardiopulmonary signals, including ECG/EKG/EMG through electrocardiographic sensors and PCG through amplified microphones or phonocardiographic transducers. Examples of commercially available devices include the 3 M Littmann CORE, which enables the recording of lung and heart sounds [25]; the Eko CORE 500, which captures both PCG and ECG data [26]; the Hexoskin Smart Shirt, which measures EKG and tidal volume [27]; and the WearME system, which records accelerometer and PCG data [28]. They support wheeze/crackle detection and respiratory rate assessment in clinical or tele-health settings. In this study, we analyze PCG signals acquired using the Littmann 3200 digital stethoscope, the AKG C417L microphone, and the Meditron electronic stethoscope [6].

3 Related Work

For detecting breathing cycles from PCG signals, published studies can be categorized according to the underlying methodology into (1) machine/deep learning techniques and (2) signal processing methods. Among the ML/DL models used for breathing cycle detection, convolutional neural networks (CNNs) [29] and long short-term memory (LSTM)

networks [30] have been explored due to their suitable capabilities in detecting peaks and cyclic patterns. Although ML and DL models are widely used in various applications today, they present certain limitations. Most DL models function as “black boxes”, making it difficult to interpret why they perform well. In addition, these models require large volumes of training data and significant computational resources, which increases system cost and limits scalability, especially for deployment on wearable devices. Moreover, Deng et al. [31] have demonstrated the importance of signal processing for feature extraction to keep the ML model simpler. The effectiveness of a ML model is also highly dependent on the quality of training data.

On the other hand, signal processing-based methods do not require prior knowledge or large amounts of annotated data, making them suitable for scenarios with limited data availability. These methods are typically deterministic and transparent, providing explainable mechanisms for detection and analysis. Moreover, signal processing techniques are computationally lightweight, making them robust and ideal for use in low-power wearable devices. Many signal processing techniques have been utilized effectively for the detection of breathing patterns from PCG signals. For instance, empirical mode decomposition (EMD) has been employed in several studies [32, 33]; spectrogram-based methods have been used in [34]; and fast Fourier transform (FFT) techniques have been applied in [35] for respiratory pattern estimation. In general, such work relies on a single technique. Our research combines multiple methods to improve detection latency and scalability.

Meanwhile, some of the published studies considered more complex signal processing algorithms that are time-consuming and, at times, lack robustness [33, 36, 37]. In addition, the bulk of the prior work focused on estimating the breathing rate rather than the cycle and conducted the analysis in the frequency rather than time domain. The reported breathing rate just reflects an aggregate view of respiratory activities and does not provide insight on how the breathing pattern varies from one cycle to another. In particular, inspiratory-to-expiratory asymmetry, the effect of pathological irregularities, and other respiratory anomalies are not fully and precisely captured by the breathing rate. Similar arguments could be made about the sole reliance of frequency domain analysis without consideration of time domain features. Table 1 provides a comparative summary of relevant work in the literature. Unlike the aforementioned studies, the proposed methods in this paper strives to determine the respiratory cycle by pursuing a two-step process, specifically: (i) estimate breathing pattern envelope (in time domain) and then (ii) use envelope peaks/valleys to estimate respiratory cycle.

4 Breathing Cycle Detection Methodology

As noted in the previous section, signal processing-based breathing cycle detection techniques in the literature either: (i) rely on a single technique or focus on respiratory rate rather than cycle, and consequently do not yield sufficiently high accuracy, or (ii) involve complex steps, raising efficiency concern. This paper promotes a different methodology, in which detecting the boundary of breathing cycles follows a two-step approach. First, key events are extracted from the raw PCG signals using a signal processing technique. These extracted features are then used in the second step to estimate the signal envelope, which represents the respiratory waveform. Respiratory signals are known to be quasi-periodic in nature [38, 39]. To effectively capture this quasi-periodicity, a combination of bandpass filtering and detection of periodic events, such as the occurrences of S_1 and S_2 peaks, is typically pursued. Envelope detection is a robust strategy for highlighting the amplitude modulation associated with respiratory cycles. Several envelope detection techniques exist, e.g., amplitude-based and Hilbert transform-based methods.

Next, a peak detection method is applied to this waveform to identify the start and end points of each respiratory cycle, enabling the detection of individual breathing cycles. We argue that such a structured approach offers the advantages of being computationally efficient and robust for both healthy and pathological signals. This section presents three methods that follow the proposed two-step process. (a) bandpass filtering with Hilbert envelope detection (BF-HED), (b) EEMD-based valley detection method, and (c) spectrogram-based envelope detection (SED) using power function analysis. The performance of these methods is evaluated using the publicly available ICBHI-2017 dataset.

4.1 Bandpass Filtering with Hilbert Envelope Detection

To detect respiratory cycles from the PCG signal, this method starts with analyzing the signal's frequency components. A bandpass filter first is used to isolate the low-frequency band where respiratory-related components are typically concentrated. Following the filtering step, the Hilbert transform is applied to obtain the signal envelope, highlighting the amplitude modulation associated with respiration. This envelope is normalized to scale the values consistently across different recordings. To identify significant respiratory peaks, we compute the 75th percentile of the normalized envelope and use it as the threshold height for peak detection. In addition, we enforce a minimum peak distance of 1 s to align with physiologically plausible breathing intervals. The detected peaks serve as indicators of respiratory cycle boundaries, enabling



Table 1 Comparison of respiratory rate vs. cycle detection approaches

Paper	Rate/cycle	Main steps/process	Envelope domain
[32]	Rate	EMD analysis; intrinsic mode function (IMF) selection based on power spectral density (PSD); fundamental frequency value detection from IMF; estimate respiratory rate through equation	No envelope
[40]	Rate	Signal filtration; Frequency band analysis; Fundamental peak detection denoted as pitch; Equation for respiratory rate as function of pitch	Frequency domain
[41]	Rate	Multiple step bandpass filtering; Welch algorithm to estimate prominent breathing frequency; Breathing rate detection by applying PSD analysis	Frequency domain
[42]	Rate	Use PPG rather than PCG; Noise filtering with high pass filter; Pulse segmentation using incremental merge segmentation algorithm; Artifact detection; Respiratory rate estimation while factoring in variations in frequency, intensity, and amplitude	No envelope
[43]	Cycle	Apply sliding window method and autocorrelation function to estimate the peak; temporary waveform generation; fine-tune the waveform by assessing the Pearson correlation coefficient; eliminate the overlapped part of the waveforms that are caused from the sliding window method	No envelope
[36]	Cycle	Bandpass filtering; Spectrogram analysis; Time domain envelope extraction by hamming window analysis; Assessing the time domain envelope by dynamic time wrapping method; Repeats until the envelope is smoothened	Time domain
[37]	Cycle then rate	Two-step entropy reduction method; breathing sound waveform detection through image shape identification; Characteristic moment waveform (breathing pattern envelope) detection; Cycle detection involves local minimum and local maximum point detection through four additional steps	Time domain
[31]	Cycle then rate	Detection of cycle by dynamic time wrapping; Peak detection; Respiratory rate detection with the peak detection; Fine-tune using Support Vector Machine	Time domain
[33]	Rate	High pass filter for noise elimination; DWT and EMD for decomposing signal components; PSD estimation of the components; SPI calculated EKF method applied to get rid of the noise; State vector fusion to obtain respiratory signal; Peak detection algorithm to estimate Breathing rate	Time domain
[39]	Rate	Bandpass filtering for noise reduction; Scalogram to identify important respiratory information; Frequency domain envelope extraction from scalogram; Respiratory Frequency extraction	Frequency domain
[35]	Rate	Noise filtering; FFT domain envelope; Peak detection	Frequency domain

estimation of breathing patterns from the PCG signal. The following provides more details.

Bandpass Filtering: The raw PCG signal contains valuable information about the heart's pumping cycle. As discussed in Sect. 2, accurate detection of the S_1 and S_2 peaks is crucial for recognizing a respiratory cycle within a PCG signal. In the frequency domain, these peaks are typically concentrated in the lower-frequency spectrum. Parmar et al. [44] report that S_1 and S_2 generally occur at frequencies below 100 Hz. Applying FFT analysis enables a more detailed investigation of the spectral characteristics of the S_1 and S_2 components.

FFT is an efficient algorithm that converts a signal from the time domain to the frequency domain. This transformation reveals the frequency components of the signal, indicating the contribution of each frequency. We apply FFT to the PCG signal sampled at 200 Hz (Nyquist rate). Figure 4 illustrates an example of FFT analysis applied to a respiratory signal. As observed in the figure, the majority of the spectral magnitude is concentrated below 10 Hz. Therefore, we select 10 Hz as the cutoff frequency for bandpass filtering. We use a fourth-order bandpass filtering method. Figure 5 illustrates



Fig. 4 Illustrating the FFT analysis of a respiratory signal

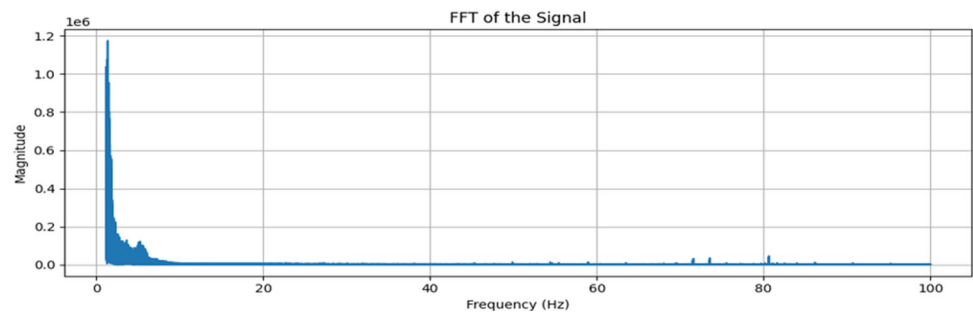
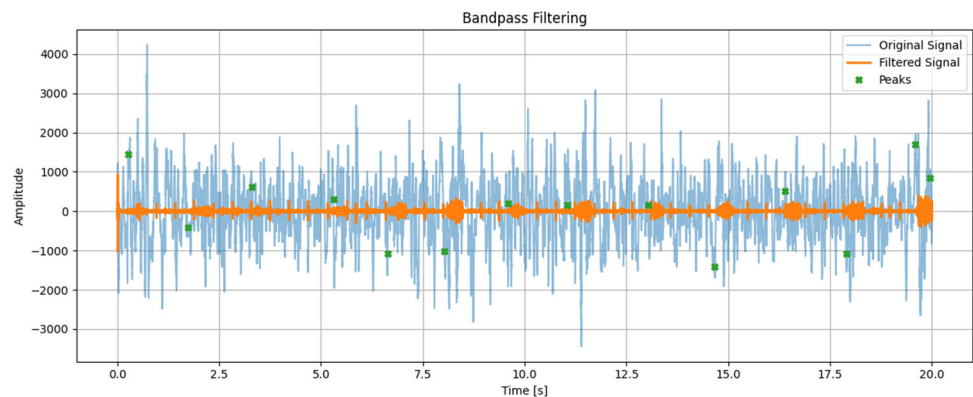


Fig. 5 An instance of the signal after bandpass filtering. The points labeled as peaks indicate the detected start and end points of the respiratory cycles in the given signal



an example of the signal after it has been processed through the bandpass filter.

Hilbert Transform: This transform is being used for envelope extraction; it has been shown to be quite effective in characterizing respiratory dynamics in previous studies [45, 46]. The Hilbert transform shifts the phase of each frequency component of a signal, $x(t)$, by -90° (or $\pi/2$ radians) for positive frequencies and $+90^\circ$ (or $-\pi/2$ radians) for negative frequencies. Mathematically, this can be defined as:

$$\mathcal{H}\{x(t)\} = \frac{1}{\pi} \text{p.v.} \int_{-\infty}^{\infty} \frac{x(\tau)}{t - \tau} d\tau \quad (1)$$

The amplitude derived from the Hilbert transform for the digital stethoscope signal is utilized to get the envelope with peaks and troughs that contain the inhalation and exhalation information [45]. Basically, the amplitude of the transformed signal smooths out the oscillations, i.e., removes the high-frequency components. Thus, we are left with the slow variations in the signal that correspond to the breathing pattern. Figure 6 shows the envelope generated by the Hilbert transform for the filtered signal in Fig. 5.

Peak detection: The detected envelope is quasi-periodic and contains multiple respiratory cycles. Each respiratory cycle includes significant peaks such as S_1 and S_2 , which exhibit notably higher amplitudes compared to the surrounding signal. Therefore, applying an appropriate peak detection method to the envelope should suffice to identify the locations of S_1 and S_2 . For accurate detection of the breathing cycle in healthy individuals, we utilize the established aver-

age respiratory rate of 12 to 20 breaths per minute [47]. Based on this rate, the duration of a single cycle can be reasonably assumed to lie within the range of 3 to 5 s.

For a healthy individual, each respiratory cycle typically consists of two phases, exhalation and inhalation. The exhalation phase generally contains the S_1 peak, while the inhalation phase usually contains the S_2 peak. These two phases commonly follow a 2:1 ratio [48], meaning that in a 3-second respiratory cycle (the minimum expected duration for a healthy person), the exhalation phase lasts approximately 2 s and the inhalation phase about 1 s. Consequently, we set the minimum peak detection distance to 1 s to ensure accurate identification of S_1 and S_2 . Another important parameter is the amplitude threshold. Prior studies [49–51] have employed percentile-based strategies for threshold selection. Inspired by these approaches, we adopt the 75th percentile of the peak amplitudes as the threshold for detecting both S_1 and S_2 peaks. Figure 7 illustrates the peak detection method applied to a sample signal. For respiratory cycle detection, among each pair of detected peaks, the lower amplitude peak is discarded, which typically corresponds to S_2 . The respiratory cycle is then defined by the elapsed time between successive S_1 peaks.

4.2 EEMD and Valley Detection

All proposed methods in this paper employ signal processing, envelope extraction, and thresholding to derive the respiratory cycle. The approach taken in this section leverages the



Fig. 6 Envelope detection with Hilbert transformation

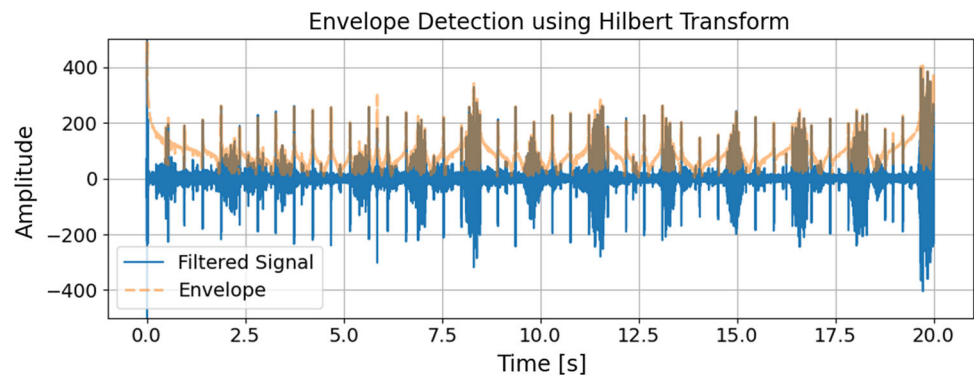
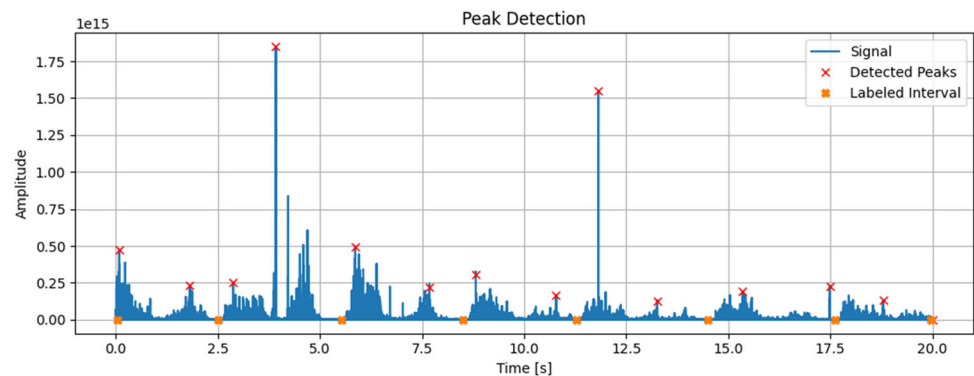


Fig. 7 Illustrating the peak detection method



capabilities of EEMD in both processing the signal to identify breathing pattern components, and extracting the envelope by selecting the most appropriate one among these components. The selected component is then subjected to valley detection to define the respiratory cycle. Details are provided below.

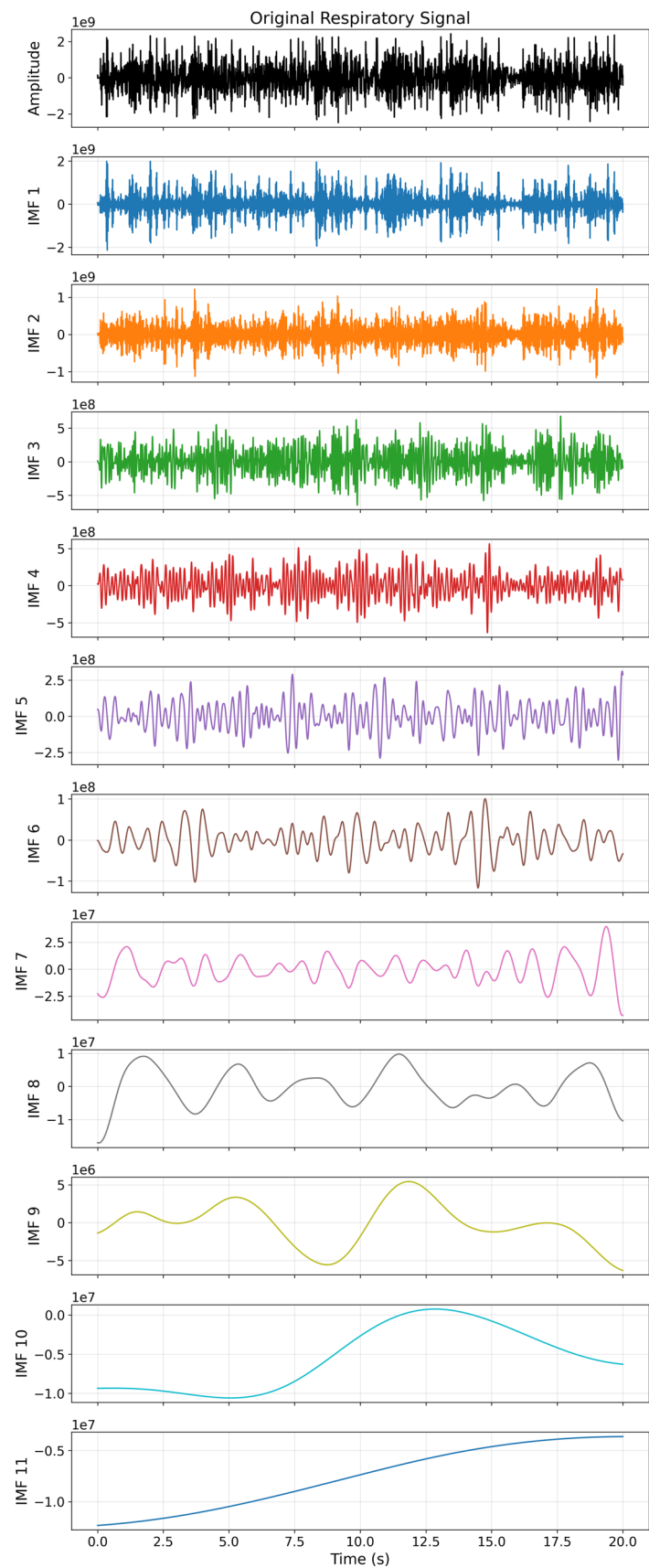
EEMD analysis: As pointed out in Sect. 4.1, it is well established that the respiratory pattern predominantly resides within the lower frequency region of the spectrum. The critical question, however, is whether FFT can efficiently capture all time-dependent frequency components of respiratory signals. The answer is partially affirmative, as FFT can represent certain classes of signals; however, it is not universally suitable for all types, particularly those exhibiting strong time-varying or non-stationary characteristics. For complex signals such as respiratory signals, FFT analysis is insufficient for capturing the inherently dynamic time–frequency variations. The empirical mode decomposition (EMD) algorithm is better suited for capturing dynamic signal characteristics. In EMD, signals are decomposed into distinct components known as intrinsic mode functions (IMFs).

For time-varying signals such as respiratory signals, a common phenomenon encountered during decomposition into IMFs is known as mode mixing. Mode mixing occurs when an IMF contains oscillations of widely different scales, or when oscillations of similar scales are distributed across multiple IMFs, typically due to signal intermittency or noise. To mitigate the mode mixing issue, an improved algorithm

known as EEMD introduces white noise into the signal, providing a uniform reference across all time scales. By averaging the results of multiple noisy trials, the random noise cancels out while the true signal components are preserved, thereby effectively reducing mode mixing. Therefore, EEMD proves to be a more effective choice compared to EMD. EEMD adaptively decomposes nonlinear and non-stationary signals into IMFs through an iterative sifting process. Iterative sifting entails identifying local extrema, constructing upper and lower envelopes and computing their mean, and subtracting such a mean from the signal. This process is repeated until the IMF conditions are satisfied, ensuring precise separation of oscillatory frequency components in complex signals [52].

The resulting IMFs are oscillatory components that possess nearly equal numbers of zero-crossings and extrema, along with a zero local mean, representing monocomponent signals at specific time scales within complex data. Each IMF captures oscillations ranging from high to low frequencies. The number of IMFs extracted through EEMD reflects the distinct oscillatory modes present in the signal, spanning from high-frequency noise to low-frequency trends. The number of IMFs depends on signal length, complexity, and noise. Each IMF represents a scale-separated component, with the final residual capturing the slowest varying baseline or trend. Although the number of IMFs may vary, the key question is which IMF should be selected to represent

Fig. 8 Showing the detected IMF by EEMD for a sample respiratory signal



breathing cycle patterns. Figure 8 shows IMF analysis of a sample chest sound signal.

IMF component selection: Respiratory information in EEMD appears in the low-frequency IMFs (0.1–0.5 Hz), while higher IMFs capture noise or cardiac activity. By selecting the IMF whose dominant frequency aligns with the respiratory signal band, one can isolate inhalation–exhalation cycles. Thus, among the various IMFs produced by EEMD, an appropriate IMF must be chosen to represent the signal envelope. Spectral flatness (SF) [53] is a prime criterion for picking an IMF that contains the breathing pattern. Spectral flatness measures how tone-like or noise-like a signal is, and is defined as the ratio of the geometric mean to the arithmetic mean of its power spectrum, i.e.,:

$$SF = \frac{\left(\prod_{k=1}^N P(k)\right)^{1/N}}{\frac{1}{N} \sum_{k=1}^N P(k)} \quad (2)$$

SF values near 1 indicate noise-like spectra, while values near 0 indicate tonal spectra.

Respiratory signals are quasi-periodic and typically confined to a narrow frequency band (0.1–0.5 Hz). In contrast, noise or mixed IMFs exhibit broadband spectral energy, which results in higher spectral flatness values. Therefore, the IMF that best represents respiration tends to show low spectral flatness, reflecting a peaked spectrum dominated by the breathing frequency. In other words, spectral flatness serves as an effective criterion for identifying IMFs that capture clear oscillatory respiratory patterns rather than noise-like components.

In this paper, we analyze respiratory signals from both healthy and unhealthy subjects. Unhealthy respiratory signals are often influenced by disease-related symptoms, such as crackles and wheezes, making them less suitable for determining which IMF component should be selected for breathing pattern analysis. As previously mentioned, the low-frequency components are typically captured in the higher-order IMFs, which can represent the breathing pattern signal. However, components below 0.1 Hz do not represent breathing patterns; rather, they correspond to noise [54]. Accordingly, the steps go as follows:

1. Get spectral flatness value of all of the IMF components of a signal using the EEMD analysis. Let us denote such a set of SF values as Γ .
2. Define the set $\gamma \subset \Gamma$ that includes the index (order) of the IMF components that have spectral flatness close to 0. The goal is to identify the IMF components that have tonal behavior resembling breathing patterns or have low-frequency noise. The rationale is that the breathing cycle is periodic and hence it would exhibit structured oscillations in the IMF.

3. Repeat steps #1 and #2 for multiple signals; this would be necessary since real life signals have varying noise patterns over time, e.g., high-frequency noises like motion noise and low-frequency noise from the collection device. Hence, some of the identified IMF components in γ actually correspond to low-frequency noise rather than the actual breathing cycle. By considering the IMF components of multiple respiratory sound signals, we opt to filter those corresponding to noise. In essence this set generates a sequence of γ sets, i.e., γ_1, γ_2 , etc.
4. Analyze the persistence of the IMF component index in γ ; since the breathing cycle will be consistently present in every considered signal, some statistical measure can be considered to filter out noise-based IMF components. Basically, the IMF component corresponding to the breathing cycle will consistently have the same index (order). Thus, we consider the index distribution and use the mode to determine the index of the breathing cycle IMF. In other words, we find the mode of the set $\bigcup_{i=1}^N \gamma_i$, where N is the number of considered signals. Note that N could be dynamically determined by repeating until the mode index becomes unique.
5. Use the mode index to select the IMF that can be treated as the signal envelope.

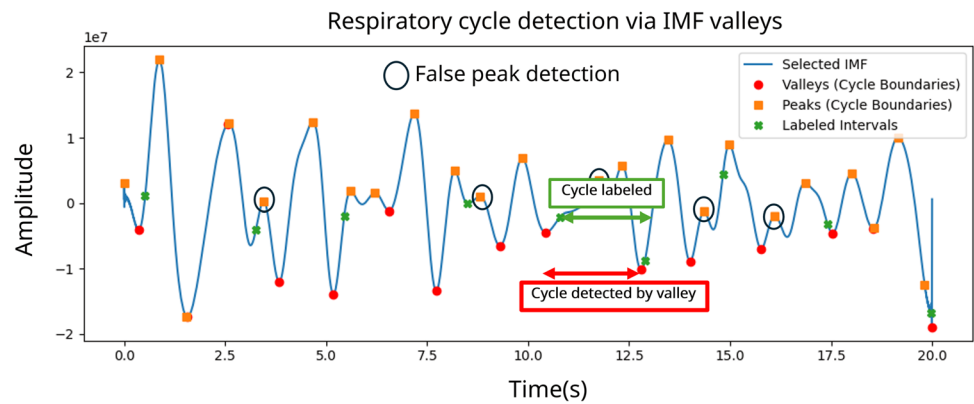
Valley detection from IMF component: Figure 8 shows the IMF envelopes, where IMF-7 corresponds to the breathing pattern. Figure 9 shows the respiratory signal envelope, where there are two peaks, one for inhalation and one for exhalation, making the respiratory cycle detection complicated. Basically, we can encounter multiple false peaks which are not reflective of one respiratory cycle. On the other hand, there is a prominent valley feature that is directly indicative of the respiratory cycle. Figure 9 contrasts peak and valley detection, whereas valleys are detected quite evidently and are coherent with the ground truth labeling. Therefore, for the IMF envelope, the valley detection is more effective than the peak detection method.

4.3 Spectrogram-Based Envelope Detection (SED)

SED uses the spectrogram for the signal processing step. The signal envelope is determined based on the spectral power intensity in the spectrogram.

Spectrogram Analysis: Both time and frequency components are crucial for analyzing the breathing sound signal. The spectrogram is a signal processing technique that can address both components simultaneously; in essence it provides a visual representation of how the frequency profile of a signal evolves over time. A spectrogram is derived from the Short-Time Fourier Transform (STFT), an extension of the Fourier Transform that enables time-localized frequency analysis. As defined by Eq. (3), STFT employs a windowing method

Fig. 9 Respiratory cycle detection with valley intervals (marked by red dots). The figure also shows the detected peaks (orange box). The encountered false peaks are marked with black hollow circles. One such instance is shown where the cycle labeled and coherent cycle detected by the valleys are indicated with green and red arrows consequently



to determine the specific time instances at which particular frequencies occur.

$$X(t, f) = \int_{-\infty}^{\infty} x(\tau) w(\tau - t) e^{-j2\pi f\tau} d\tau \quad (3)$$

where $x(\tau)$ is the input signal, $w(\tau - t)$ is a window function centered at time t , and f represents frequency. The spectrogram is then defined as the squared magnitude of the STFT:

$$S(t, f) = |X(t, f)|^2 \quad (4)$$

The key advantage of STFT is its ability to incorporate both time and frequency information, which in spectrogram analysis can be simultaneously used as parameters. By extracting the spectrogram $S(t, f)$ as a function of time (t) and frequency (f), the resulting values can be utilized to calculate the envelope. Figure 10 illustrates the spectrogram analysis of a sample digital stethoscope recording. A regular pattern of spectral power can be observed, which can be further utilized for envelope detection, as explained next.

Envelope Determination: A spectrogram provides the spectral power intensity for all frequencies at each time instance. In Fig. 10, for low frequencies (the lower part of the y-axis), the power intensity exhibits a quasi-periodic pattern which reflects the prominent breathing pattern component powers. For certain time instances, e.g., at 15 sec and 18 sec in the figure, we can see intense power for the [0–10,000] Hz frequency range. These two instances signifies the importance of considering the entire spectral range since high power density can randomly appear within the spectrogram. However, mostly significant power components can be observed in the lower frequency range when considering respiratory signals. We leverage these aspects of the spectrogram and generate the envelope; in other words, time, frequency and power information are used to determine the signal envelope.

In the context of respiratory sound signals, the spectral power is computed as functions of both time and frequency. As stated earlier, the breathing pattern information is usually contained within the low-frequency range. However, the

corresponding power values are much lower and are suppressed by the high-frequency components, which primarily represent noise. An additional consideration is that pathological signals often exhibit periodic instances of adventitious sounds such as wheezes and crackles. Therefore, capturing the high-frequency power components is essential for the characterization of unhealthy respiratory signals; the formulation can be expressed as:

$$E(t) = S(t, f) \left(\alpha + \beta \cdot \frac{1}{(f + f_e)} \right), \quad (\alpha, \beta) \neq (0, 0) \quad (5)$$

where β accentuates the low-frequency power by scaling the spectral power with the inverse of frequency, while α captures the global power distribution across the entire frequency spectrum. In the experimental analysis, the parameters α and β are varied to illustrate their impact on breathing cycle pattern detection.

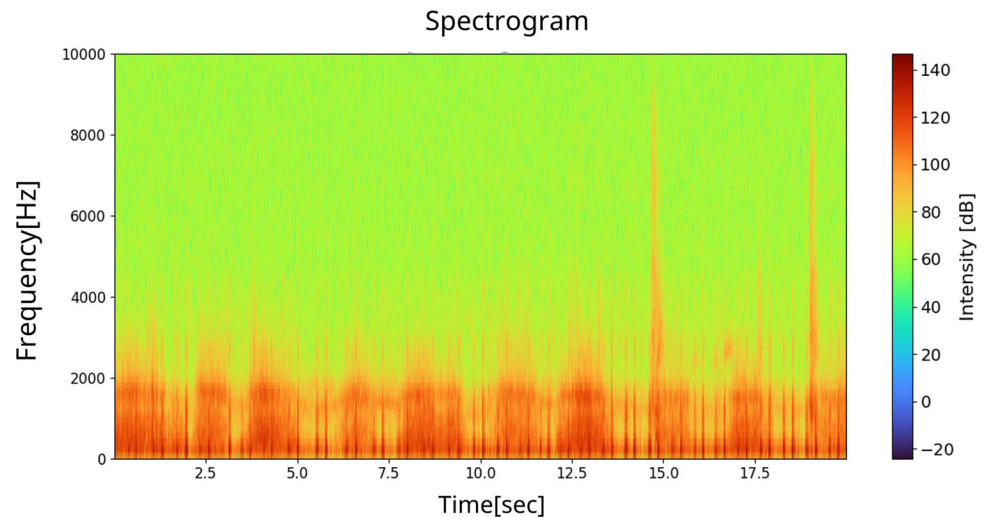
5 Performance Validation

5.1 Dataset and Performance Metrics

We utilize the ICBHI-2017 Respiratory Sound Database [6] for assessing the effectiveness of the proposed methods. The dataset has become a standardized benchmark for the automatic analysis of lung sounds, with a primary emphasis on respiratory disease detection and classification. The dataset consists of 920 recordings obtained from 126 subjects, encompassing both patients and healthy volunteers; it includes recordings from subjects diagnosed with COPD, asthma, bronchiectasis, and lower or upper respiratory tract infections, along with recordings from healthy individuals. The recordings were collected from the trachea, anterior (left and right), posterior (left and right), and lateral (left and right) lung fields. This systematic selection of recording positions captures both central and peripheral respiratory sounds,



Fig. 10 Spectrogram analysis of a sample respiratory signal



thereby enhancing the diagnostic diversity of the dataset and enabling more robust analyses of region-specific acoustic patterns.

The ICBHI dataset has been collected using varieties of digital stethoscope like AKG C417L Microphone, 3M Littman Classic II SE Stethoscope, 3M Littmann 3200 Electronic Stethoscope, WelchAllyn Meditron Master Elite Electronic Stethoscope, etc. The samples were collected at multiple hospitals throughout the years. Specifically, the recordings are made independently by two teams working in different countries. Most samples were captured by ESSUA at Lab3R and Hospital Infante D. Pedro in Aveiro, Portugal, while the AUTH-UC team collected additional data in hospitals in Thessaloniki and Naousa, Greece. The breathing cycle for each sample was annotated by medical experts in these hospitals, starting the respiratory cycle at the onset of inspiration and ending it when expiration was complete.

Some of the unhealthy signals in the dataset are so distorted with heavy cough and anomaly sounds that extensive and sophisticated filtering methods will be required. Since the focus is on validating the two-step methodology, unhealthy signals that could not be effectively filtered using the first step are not considered. Subsequently, we ended up selecting 24 healthy lung sound samples and 36 unhealthy lung sound samples. The signals are from 29 unhealthy individuals and 22 healthy individuals, the former encompasses a variety of respiratory diseases. Figure 11 illustrates the wide variability in respiratory cycle lengths for both healthy and unhealthy signals in the considered subset.

Given the design objective, the performance of the proposed algorithms is evaluated based on their accuracy in detecting the respiratory cycles. A set of error estimation metrics are used, comparing the actual cycle length, y_i , with the predicted cycle length, $\hat{y}_i$, averaged over N respiratory cycles. Specifically, we use:

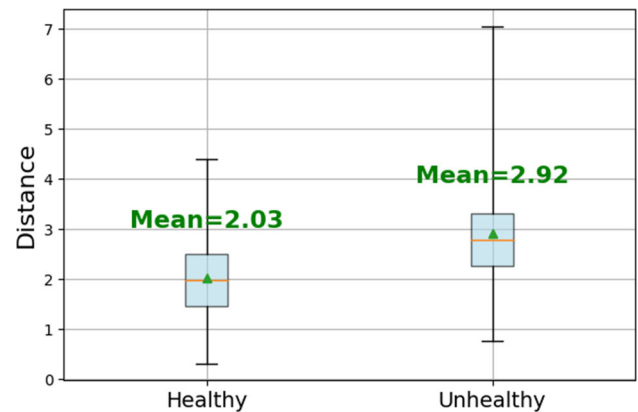


Fig. 11 The mean breathing cycle length for healthy and unhealthy subjects in the considered subset of ICBHI-2017 dataset

- **Mean Absolute Error (MAE):** MAE is a statistical measure reflecting the average magnitude of errors between predicted values and reference values. It is defined as:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (6)$$

where $|\cdot|$ indicates the absolute value. MAE treats all errors linearly, which is particularly important in respiratory signal analysis where occasional irregular breaths, motion artifacts, or coughs may occur. Unlike error metrics that disproportionately emphasize large deviations, MAE provides a stable and reliable performance estimate without being overly influenced by such outliers. In this context, MAE serves as an appropriate measure, as it yields a direct and interpretable estimate of the average error magnitude in physiologically meaningful units (seconds per cycle) [55]. This property makes MAE especially valuable for clinical evaluation, where under-

standing the typical deviation from the true cycle length is critical for assessing whether a system is sufficiently accurate for diagnostic or therapeutic decision-making.

- **Root Mean Square Error (RMSE):** RMSE calculates the square root of the average squared differences between predicted and true values, making it a sensitive metric that emphasizes larger errors.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (7)$$

The respiratory signal is continuous in nature, with breathing cycles occurring in a successive and uninterrupted manner. While overestimating or underestimating the cycle length of a single respiratory period may not seem critical, such an error can accumulate, and if the length of subsequent cycles is miscalculated, the overall error can become substantially larger. As indicated by Eq. (7), RMSE places greater emphasis on errors, making the overall result more strongly influenced by erroneous calculations than by accurately estimated cycle lengths. Thus, RMSE is an effective metric for comparing the performance of respiratory cycle detection algorithms.

- **Mean Absolute Percentage Error (MAPE):** MAPE measures the average absolute difference between predicted and true values, expressed as a percentage of the true values. In essence, MAPE provides a normalized indicator of prediction accuracy. This distinction is especially important in our study, since the dataset encompasses both healthy and unhealthy respiratory cases. In unhealthy cases, respiratory cycles may be abnormally short or prolonged, and hence relying solely on absolute units may not accurately reflect the algorithm's performance. Normalizing the error places both healthy and unhealthy respiratory cycles on the same scale, allowing them to be evaluated on equal terms. Mathematically, MAPE can be calculated as:

$$\text{MAPE} = \frac{100}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (8)$$

From the equation, it can be inferred that MAPE is a relative measure that normalizes the error with respect to the true value.

5.2 Performance Results

Although the respiratory sound signal is generally quasi-periodic, in practice it often gets distorted due to variations in the human activities, instrumentation error, and ambient noise. Our methods mitigate these issues by: (1) pursuing a two-step process that first gets rid of the noise and then detect

the signal envelope, and (2) factoring both time and frequency domain attributes, defined from the knowledge of respiratory signals (data informed decision). This section validates these capabilities. The performance experiments have focused on: (1) assessing the accuracy of detecting breathing cycles in the ICBHI-2017 dataset [6], highlighting the advantages of the proposed methods over competing approaches in the literature, and (2) comparing the proposed methods to derive some guidelines for practical use.

5.2.1 Comparison with Existing Methods

We compare the performance with three baseline approaches from the literature, employing bandpass filters [41], EMD [56], or cepstrum [57]. The results of the proposed methods and the competing approaches are summarized in Tables 2 and 3. For SED, we study the effect of the parameter setting in Eq. (5) and consider two variants: (i) SED_s , where $\alpha = 1$, and $\beta = 0$, and (ii) SED_f , corresponding to the setting $\alpha = 0$, and $\beta = 1$. The three baseline approaches strive to estimate the respiratory rate (RR) rather than the breathing cycle. Although RR estimation is a widely used clinical parameter, it provides only a rough summary of respiratory activity. As noted in Sect. 3, RR reduces the signal to an average frequency measure, thus overlooking cycle-to-cycle variability, inspiratory-to-expiratory asymmetry, amplitude dynamics, and pathological irregularities. Therefore, relying on RR alone cannot comprehensively capture respiratory anomalies. Such a limitation is highlighted by the error evaluation results in Figs. 12, 13, and 14, which show the variations in the duration of the breathing cycle when factoring in only RR, even for healthy individuals. This is in contrast to the robust results achieved by the proposed methods.

Bandpass filter based methods: We highlight the effectiveness of our proposed BF-HED method by comparing its performance with an approach that solely relies on a bandpass filter. In particular, we consider the approach of [41], which estimates the respiratory rate from tracheal or nasal breath sounds recorded at 44.1 kHz by first applying a 500–5000 Hz bandpass filter and extracting the necessary respiratory sound signal, $r(t)$, by using Hilbert transform. $r(t)$ is digitized at 100 Hz and resampled through cubic spline interpolation to ensure uniform sampling to achieve $r[n]$. A secondary bandpass filter (0.19–4.6 Hz) isolates the respiratory band, $r_b(t)$, followed by signal cropping and down-sampling to 10 Hz for efficient processing. Power spectral density is then calculated, and peak detection is performed to identify the dominant periodic components, from which the respiratory rate is calculated.

For the considered dataset, the baseline approach failed to detect the breathing cycles at rate of 14.28% and 12.5% for healthy and unhealthy cases, respectively. Meanwhile, BF-HED successfully processed all the cases. The other met-



Table 2 The performance of the proposed methods for sound signals of healthy and unhealthy individuals in the considered dataset

Method	MAE (Average)	RMSE (Average)	MAPE (Average)
BF-HED_Healthy	0.5650	0.7049	34.1244
BF-HED_Unhealthy	0.7749	0.9922	27.2530
EEMD_Healthy	0.7024	0.8279	36.6435
EEMD_Unhealthy	0.9993	1.1518	34.7559
SED_s_Healthy	0.9906	1.1714	54.4970
SED_s_Unhealthy	0.9390	1.2144	30.8000
SED_f_Healthy	0.9396	1.1076	49.5919
SED_f_Unhealthy	1.1776	1.3870	34.7090

Table 3 How the baseline methods Perform on healthy and unhealthy signals

Method	MAE (Average)	RMSE (Average)	MAPE (Average)
Bandpass_Healthy [41]	10.4780	10.4780	37.0016
Bandpass_Unhealthy [41]	5.2664	5.2664	27.1107
EMD_Healthy [56]	10.5740	10.5740	32.8989
EMD_Unhealthy [56]	6.1799	6.1799	33.1190
Cepstrum_Healthy [57]	2.1191	2.8106	91.0618
Cepstrum_Unhealthy [57]	2.4357	3.0313	51.9873

rics, MAE, RMSE, and MAPE, also widely varied between the two methods, where BF-HED could consistently yield more accurate estimates, confirming its robustness and ability to handle dynamic fluctuations in the length of respiratory cycles.

EEMD vs. EMD: A respiratory signal that is acquired by a digital stethoscope, is a nonlinear and non-stationary noisy signal. While EMD is an effective method to denoise the signal, the breathing pattern from the monocomponent analysis, in this case, IMF acquisition, degrades such denoising ability due to mode mixing, as discussed in Sect. 4.2. The proposed EEMD method mitigates this shortcoming. To validate our hypothesis, we compare EEMD with the method described in [56]. The breathing pattern analysis results in Figs. 12, 13, 14 clearly confirm that EEMD outperforms EMD. As indicated by the results in Tables 2 and 3 which compare the MAE and RMSE average, EEMD performs 14 and 6 times better than EMD for healthy and unhealthy signals, respectively.

To highlight the effect of the IMF decomposition on the performance, we have calculated the orthogonality index (OI). While IMFs are being compartmentalized, they should not depend on each other, i.e., the IMFs should be orthogonal. The OI between the j^{th} IMF and the k^{th} IMF can be expressed as [58]:

$$OI_{jk} = \frac{\int_0^T c_j(t) c_k(t) dt}{\sqrt{\int_0^T c_j^2(t) dt \int_0^T c_k^2(t) dt}} \quad (9)$$

In an ideal case, IMF_j and IMF_k are orthogonal and OI_{jk} has a value of zero. When comparing the IMFs of EEMD and EMD, the orthogonality index was found to be on average 71% lower for EEMD than EMD, which demonstrates EEMD's ability to resolve the mode mixing issue.

Spectrogram vs. Cepstrum: A spectrogram recognizes all anomalies and periodicity patterns of the respiratory signal. A cepstrum, on the other hand, is an efficient algorithm for analyzing the components that are predominant in the signal; however, a cepstrum fails to capture the transient nature of the variable respiratory signal. Thus, for unhealthy respiratory signals, the cepstrum does not perform well in capturing the envelope of a breathing signal, a shortcoming that SED avoids. With the help of Eq. (5), SED utilizes the spectrogram capability of extracting information throughout the frequency and time domains. To validate the advantage of SED, we compare its performance with the cepstrum-based method of [57]. Such a baseline method first passes the signal through a bandpass filter before being processed with a cepstrum. The cepstrum equation is:

$$C_R = \left| \mathcal{F}^{-1} \left\{ \log \left(|\mathcal{F}\{R(n)\}|^2 \right) \right\} \right|^2 \quad (10)$$

where, FFT is first applied to the signal, the logarithm is taken, and then an inverse FFT is performed. This provides the frequency values where repeated patterns, i.e., tone-like signals, are evident, in our case the breathing frequency.

The results in Figs. 12, 13, and 14 clearly confirm the performance edge that SED has over the baseline cepstrum method. As discussed in Sect. 4, the respiratory signal is not

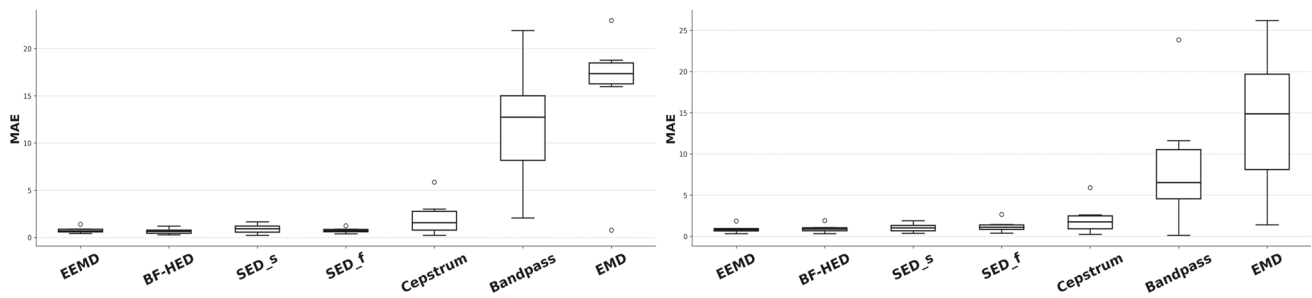


Fig. 12 The observed MAE results, separated based on whether the signals correspond to healthy(left) and unhealthy(right) individuals. The small circle represents values that are not in the range of the box plot

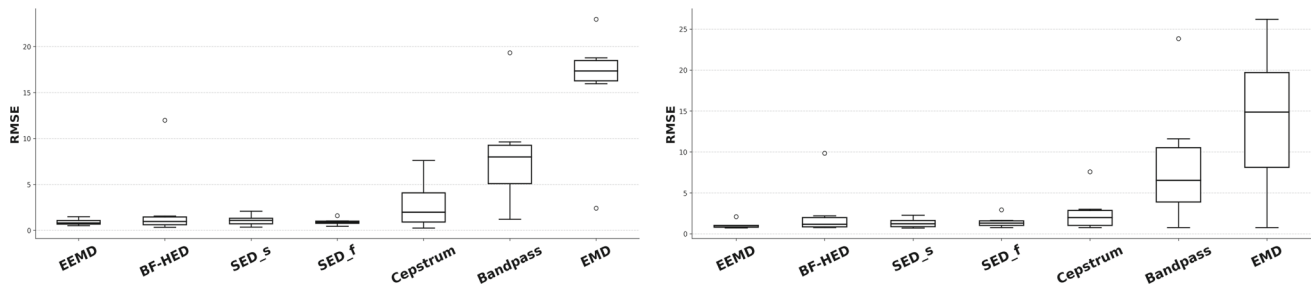


Fig. 13 The RMSE results for healthy (left) and unhealthy (right) cases. The small circle reflects values that are not in the range of the box plot

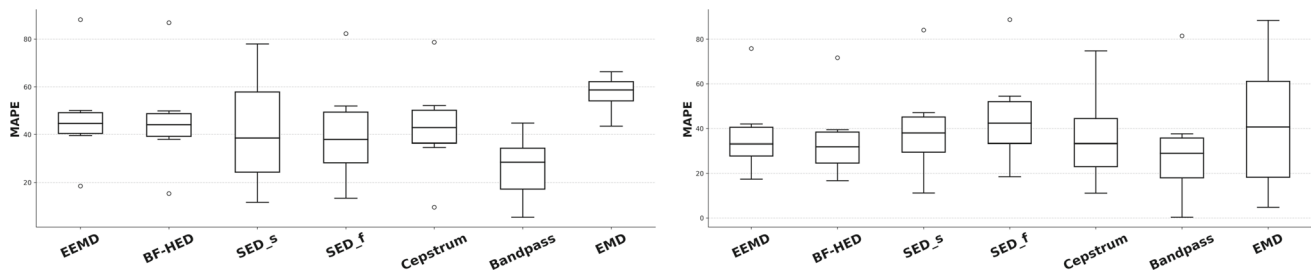


Fig. 14 Comparing the results for healthy (left) and unhealthy (right) cases under the MAPE metric. The small circle represents values that are not in the range of the box plot

entirely periodic and irregular events are encountered, especially under unhealthy conditions. In this situation, a method that factors in time and frequency domain information, like SED, will be more able to capture the dynamic variation encountered in the frequency through different time frames. A cepstrum over-processes the information from the signal causing it to miss some of the instantaneous changes in the signal.

5.3 Guidelines for Method Selection

The results reported so far demonstrate the performance advantages of our proposed methods in terms of respiratory cycle estimation accuracy. The remaining question is how to rank the three proposed methods. To answer this question, we consider two more metrics, namely, the time complexity and the quality of the detected signal envelope.

Computational Complexity: The complexity of the proposed methods is evaluated using a Raspberry pi 4 platform. The observed runtime of all methods is given in Fig. 15. We also report the runtime for the baseline approaches as well. Among the proposed two-step methods, BF-HED is the most time efficient. As discussed in Sect. 4.1, BF-HED employs a bandpass filter and Hilbert transform for the respiratory envelope detection which is more lightweight than the EEMD and Spectrogram analysis.

To assess memory usage, we evaluate the resident set size (RSS) of each algorithm. RSS measures the amount of physical RAM actively used by a running program and serves as a practical indicator of memory efficiency. Lower RSS indicates reduced use of in-memory data structures and intermediate buffers, which is critical for resource-constrained platforms, where excessive memory usage can degrade performance. Algorithms with lower RSS also scale better with signal length, support stable real-time execution, and are



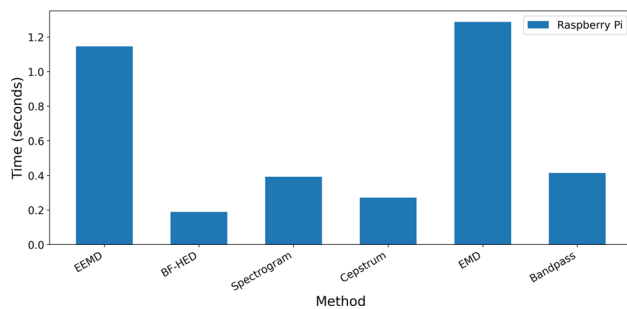


Fig. 15 Time complexity comparison for the proposed and baseline methods

Table 4 Memory usage by algorithms on Raspberry Pi 4

Method	Raspberry Pi (MB)
EEMD	5.465
BF-HED	1.015
Spectrogram	2.253
Cepstrum	2.10
EMD	6.62
Bandpass	1.105

more portable across hardware platforms. The RSS measured on Raspberry Pi device is reported in Table 4, confirming efficiency and feasibility.

Envelope quality: As discussed, we are following a two-step process, where the first step involves detecting the breathing signal envelope and the second step involves the respiratory cycle detection from the envelope. Therefore, the envelope detection accuracy is very critical. To assess the envelope quality of our proposed methods, we have used Pearson correlation. In statistics, the Pearson correlation coefficient (PCC) is a measure of the strength and direction of the linear relationship between two continuous signals. In the context of our methods, we consider the coefficient value, c , and the observed correlation's statistical significance, p . For c , a value closer to 1 indicates that the envelope is well-related to the respiratory signal acquired from the digital stethoscope, while the p value is desired to be less than 0.05. Table 5 reports the envelope quality assessment results for both healthy and unhealthy individuals. The results show that the EEMD method yields a good quality envelope in both cases. However, the Spectrogram method is superior for unhealthy cases.

Method Recommendation: Among the proposed three methods, BF-HED is the fastest according to Fig. 15 with an execution time of 0.179 sec. Thus, BF-HED is efficient for conducting the respiratory cycle analysis in real time compared to the two methods. The S_1 and S_2 peaks (the lub-dub sounds) are prominent features in digital stethoscope signals that doctors use to assess the respiratory health. BF-HED can

Table 5 Pearson correlation coefficient (c) and p -values for the actual respiratory signals and detected envelopes for healthy and unhealthy cases

Method	Healthy (c , p)	Unhealthy (c , p)
EEMD	0.68, 3.06×10^{-307}	0.35, 1.04×10^{-210}
Spect (Sxx)	0.15, 0.01	0.40, 0.04
BF-HED	0.24, 5.6×10^{-38}	0.27, 1.28×10^{-34}
Spect (Sxx f)	0.30, $6e-3$	0.24, 0.02

be utilized for monitoring of diseases, such as COPD and asthma, where the lub-dub sounds are frequent than usual and rapid processing is required.

Meanwhile, EEMD yields the best envelope quality according to Table 5 with the envelope having a Pearson correlation of 0.68 and 0.35 for healthy and unhealthy cases, respectively. Inspecting the envelope from Fig. 9, we can see an ideal envelope representation. EEMD also has the best MAE, RMSE score for the cycle detection according to Figs. 12 and 13. The envelope quality itself is reflective of other parameters that can be indicative of respiratory disease, where the lub-dub sound detection does not suffice. For example, for COPD, the length of the inhalation cycle, exhalation cycle and the pauses between these two cycles is insightful for determining COPD severity. Thus, EEMD will be handy in these instances.

Using a spectrogram has the option of tweaking the α and β parameters to suit the application, as reflected in Eq. (5). The setting $\alpha = 1$ and $\beta = 0$, referred to as SED_s , performs the best for cycle detection of unhealthy signal instances with an average MAE, RMSE and MAPE of 0.939, 1.214 and 30.799, respectively. Table 5 explains such performance, where the Pearson correlation of the envelope is the maximum for unhealthy conditions. Meanwhile, SED_f has the least variation under the MAE, RMSE and MAPE metrics with an average of 0.939, 1.171 and 54.497, respectively (Figs. 12, 13 and 14). Thus, the spectrogram-based method is the most robust for breathing cycle detection.

6 Conclusion

Estimating the respiratory cycle is instrumental for detecting breathing abnormalities. Tele-health applications rely on the use of acoustic sensors and automating the cycle calculation. The paper has focused on the use of signal processing techniques to determine the breathing cycle within a chest sound recording. Most published methods are generally complex methods and focus on breathing rate rather than cycle. This paper argues that the respiratory rate does not capture the dynamic variation encountered in the breathing signals. This paper fills the gap by promoting a two-step process to extract

the envelope of the breathing sound signal and then determine the respiratory cycle from the envelope using a peak and valley detection strategy. Three methods are presented to highlight the utility of such a process, using bandpass filters, EEMD and spectrogram. The validation results using a public dataset have demonstrated the more accurate on the basis of metrics like MAE, RMSE and MAPE and faster with respect to time complexity analysis. In the future, we plan to utilize the proposed methods to further analyze the inhalation and exhalation cycle parameters to diagnose several respiratory diseases such as COPD, pneumonia, and asthma.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the scientific findings reported in this paper.

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